

S2.2.

P3. solution:

$$2x - 4y = 6 \quad \text{--- (1)}$$

$$-x + 5y = 0 \quad \text{--- (2)}$$

$$\text{multiplier: } \frac{-1}{2}$$

$$\text{(2)} - (-\frac{1}{2}) \times \text{(1)} \Rightarrow$$

$$\begin{cases} 2x - 4y = 6 \\ 3y = 3 \end{cases}$$

$$\Rightarrow y = 1, x = 5$$

When the right side changes to $(-6, 0)$, the new solution is $x = -5, y = -1$

P6. solution:

$$2x + by = 16 \quad \text{--- (1)}$$

$$4x + 8y = 9 \quad \text{--- (2)}$$

$$\text{multiplier: } \frac{4}{2} = 2$$

$$\text{(2)} - 2 \times \text{(1)} \Rightarrow$$

$$(8 - 2b)y = 9 - 32$$

$\Rightarrow 8 - 2b = 0$ that will make this system singular

$$\Rightarrow b = 4$$

To make it solvable, we need

$$9 - 32 = 0$$

$$9 = 32$$

The final system is

$$2x + 4y = 16$$

$$4x + 8y = 32$$

Two solutions: i) $x = 0, y = 4$

ii) $x = 8, y = 0$

P8. solution:

$$kx + 3y = 6$$

$$3x + ky = -6$$

$$\text{i) } k = 0$$

$$\begin{cases} 3y = 6 \\ 3x = -6 \end{cases} \rightarrow \text{fail}$$

This case can be fixed by a row exchange and it has a unique solution $x = -2, y = 2$

ii) When $k \neq 0$, we have a multiplier $\frac{3}{k}$, then

$$\begin{cases} kx + 3y = 6 \\ (k - \frac{9}{k})y = -6 - \frac{18}{k} \end{cases}$$

$$k - \frac{9}{k} = 0 \rightarrow \text{fail}$$

$$\Rightarrow k^2 - 9 = 0$$

$$\Rightarrow k = 3 \text{ or } -3$$

When $k = 3$, $0 \cdot y = -12$, this system has no solution

When $k = -3$, $0 \cdot y = 0$, this system has infinitely many solutions

P15. solution:

$$\text{(a) } y + z = 2$$

$$z = 1 \Leftrightarrow [A \ b] = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{bmatrix}$$

$$x + y + z = 3$$

$$P_{33} P_{13} [A \ b] = P_{33} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 1, z = 1$$

$$\text{(b) } y + z = 2$$

$$x + y + z = 3 \Leftrightarrow [A \ b] = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

$$y + z = 3$$

$$E_{32} P_{12} [A \ b] = E_{32} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{with } E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

fail.

S2.3

P1. Solution:

$$(a) E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(b) E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$$

$$(c) P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P = P_{23} P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

P3. Solution:

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$M = E_{32} E_{31} E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix}$$

$$MA = U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

P9. Solution:

$$(a) E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M = P_{23} E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$(b) P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$M = E_{31} P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

P12. Solution:

$$(a) \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 2 & -3 \end{bmatrix}$$

P25. Solution:

$$[A \ b] = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 3 & 4 & 2 \\ 3 & 5 & 7 & 6 \end{bmatrix}$$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$$E_{31} E_{21} [A \ b] = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -2 & 0 \\ 0 & -1 & -2 & 3 \end{bmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$E_{32} E_{31} E_{21} [A \ b] = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

The last row means

$$0.x + 0.y + 0.z = 3 \rightarrow \text{impossible}$$

$\Rightarrow$ This system has no solution.

The last number 6 should be changed to 3, then this system has infinitely many solutions.

P28. Proof: $AB = I \Rightarrow (AB)C = I \cdot C = C$

$$BC = I \Rightarrow A(BC) = A \cdot I = A$$

$$\Rightarrow C = A \text{ Since } (AB)C = A(BC)$$

S2.4

P1. Solution:

$$BA = \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 5 & 5 & 5 \\ 5 & 5 & 5 \\ 5 & 5 & 5 \end{bmatrix}$$

$$ABD = \begin{bmatrix} 15 \\ 15 \\ 15 \end{bmatrix}$$

$DBA, A(B+C)$
not allowed.

$$P5. i) A = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2b \\ 0 & 1 \end{bmatrix}, A^3 = \begin{bmatrix} 1 & 3b \\ 0 & 1 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 1 & nb \\ 0 & 1 \end{bmatrix}$$

$$ii) A = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix}, A^3 = \begin{bmatrix} 8 & 8 \\ 0 & 0 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 2^n & 2^n \\ 0 & 0 \end{bmatrix}$$

P7. Solution:

a) True

b) false

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

c) True

d) false.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B^2 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} = (AB)^2 \neq A^2 B^2 = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$P11. a) B = 4I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$b) B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$cc) B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$d) B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

P23. Solution: $\rightarrow$ or $\begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$

$$i) A = \begin{bmatrix} \pm i & \\ & \pm i \end{bmatrix}, A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$ii) B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BC = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$iii) D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, E = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$\left(\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} = DE \right) = \left(ED = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \right)$$

P25. Solution:

$$i) A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, A_1^2 = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}, A_1^3 = \begin{bmatrix} 8 & 7 \\ 0 & 1 \end{bmatrix}$$

$$A_1^n = \begin{bmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{bmatrix}$$

$$ii) A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, A_2^2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}, A_2^3 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$$

$$A_2^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$$

$$iii) A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}, A_3^2 = \begin{bmatrix} a^2 & ab \\ 0 & 0 \end{bmatrix}, A_3^3 = \begin{bmatrix} a^3 & a^2 b \\ 0 & 0 \end{bmatrix}$$

$$A_3^n = \begin{bmatrix} a^n & a^{n-1} b \\ 0 & 0 \end{bmatrix}$$